# Sean Luka Girgis


Python / AWS Developer | Financial Services, SDLC, Testing & Production Support

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LinkedIn | GitHub | Portfolio


## Professional Summary


Senior Python, AWS, and financial-services technology professional with 20+ years of enterprise experience across Python, SQL, AWS data/platform workflows, Oracle/MySQL concepts, SDLC delivery, requirements analysis, design documentation, testing, production support, troubleshooting, REST/SOAP integration concepts, and AI-assisted development. Strong background translating business and technical needs into reliable software/data workflows, documenting designs and support procedures, validating outputs, resolving production issues, and collaborating with cross-functional teams in regulated financial-services environments. Best aligned to Python/AWS consulting roles requiring hands-on development, SDLC discipline, testing, code review concepts, database development, production reliability, documentation, and client/stakeholder communication; modern web technologies, Java, JavaScript, and broad full-stack application ownership are positioned carefully as adjacent or earlier-career/working-knowledge areas rather than overstated current production ownership.


## Professional Experience

**Senior Capacity & Data Engineer at CITI** (2017-11 – 2025-12)

- Built and supported Python/Pandas/PySpark and SQL workflows ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for reporting, analytics, forecasting, validation, and operational decision support.
- Developed AWS and warehouse-backed platform components using S3, AWS Glue, Athena, Amazon Redshift, Oracle, SQL, and Python/PySpark patterns to support scalable ingestion, transformation, querying, and analytics.
- Supported end-to-end SDLC-style delivery by clarifying requirements, designing data flows, implementing code changes, validating outputs, documenting behavior, and coordinating fixes with application, infrastructure, and business teams.
- Troubleshot production issues by tracing source feeds, Python processing, SQL transformations, metric behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Implemented testing and validation checks for row counts, nulls, duplicates, unmatched joins, data freshness, abnormal deltas, and output comparisons against trusted prior runs.
- Created runbooks, source mappings, conceptual data flows, metric definitions, validation checkpoints, support procedures, and technical documentation to enrich team knowledge bases and improve handoff.
- Used AI-assisted development tools including ChatGPT/Copilot-style workflows and GenAI/RAG-oriented project work to accelerate documentation, code review thinking, test generation, and data workflow development.
- Communicated technical and operational issues clearly to technical and non-technical stakeholders in a regulated financial-services environment.

**Performance Engineer at G6 Hospitality LLC** (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Analyzed application and monitoring data to identify production-like performance issues, reliability risks, data quality issues, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical system improvements.

**SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF** (2011 – 2016)

- Supported a large financial-services monitoring and reporting environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing and support workflows.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

**Performance Test Engineer at AT&T** (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed J2EE application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during testing and production-readiness reviews.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

**Senior Systems & Data Migration Engineer at Sabre** (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported schema conversion, data validation, phased cutover, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

**Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC)** (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, VC++ components, and DB2-backed integration modules in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

**Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint** (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using C, C++, Pro*C, Oracle, PL/SQL, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, and release support for telecom billing and customer-management applications.
- Maintained code in CVS and ClearCase and supported legacy application environments across HPUX, SunOS, Linux, WebLogic, WebSphere, Oracle, and messaging interfaces.

## Education

**Post-Graduate Diploma in Computer Engineering Technology / Computer Science**
Humber College, Toronto, Canada

**Bachelor of Science in Civil Engineering**
Zagazig University, Egypt

## Skills

## Flagship Projects

**Serverless Lakehouse Platform (AWS)**
Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, curated transformations, analytics datasets, query optimization, and validation checkpoints.
**Technologies:** AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

**ServiceCall AI RAG Demo**
Built staged Python proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, Pydantic schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.
**Technologies:** Python, FastAPI, Docker, Pydantic, pytest, RAG, TF-IDF

**HorizonScale — Capacity Forecasting and Data Platform**
Built Python/Pandas/PySpark workflows to process telemetry, validate outputs, prepare forecasting datasets, and produce trusted capacity and risk reporting for financial-services infrastructure.
**Technologies:** Python, Pandas, PySpark, SQL, Prophet, scikit-learn, Data Validation

**Enterprise Application Monitoring and Support**
Supported production monitoring, troubleshooting, dashboarding, and documentation across CA APM, Dynatrace, AppDynamics, JMX, and financial-services enterprise applications.
**Technologies:** CA APM, Dynatrace, AppDynamics, JMX, Documentation, Troubleshooting